

Linzan Ye

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EDUCATION

Carnegie Mellon University

M.S. in Music and Technology

Pittsburgh, Pennsylvania

Aug. 2025 – May 2027 (expected)

- Research interests: online inference, language modeling, music information retrieval
- Awarded a merit-based fellowship covering 50% of tuition

University of Rochester

B.A. in Music (Highest Distinction) & B.A. in Data Science (High Distinction)

Rochester, New York

Aug. 2017 – May 2021

- Cumulative GPA: 3.92 out of 4.0; Dean's List (all eligible semesters)

RESEARCH EXPERIENCE

Carnegie Mellon University

Pittsburgh, Pennsylvania

Advisors: Roger Dannenberg and Chris Donahue

My research focuses on augmenting real-time musical interactions through controllable intelligent systems capable of understanding and responding to multimodal human input with low latency. My work combines real-time music understanding and generation, representation learning, and online inference.

Real-time Performance-MIDI-to-Score with Latent Distillation

Aug. 2025 – Present

- Formulated strictly causal, event-synchronous performance-MIDI-to-score transcription and trained a decoder-only model that emits a compact score token at each note onset within a 10 ms latency budget on consumer hardware.
- Proposed note-aligned latent distillation from a heterogeneous encoder-decoder teacher, reducing MUSTER error by 6.2% relative to the non-distilled model.
- Designed a hybrid framework where offline predictions periodically refine the score and causal decoding history without representation translation, improving the online model's subsequent onset and staff predictions.

PiCoMaestro: An Interactive Interface for Piano Conducting

Jun. 2026 – Present

- Developed a web-based interface for real-time piano performance that lets users control the articulation, velocity, and pedaling of the melody while a rule-based algorithm generates accompaniment from users' expression. Curated a dataset of 165 melody-annotated piano scores for music information retrieval.
- Built an automated, consent-based pipeline to collect user performance data for expressive piano performance modeling, score following, and automatic accompaniment.
- Designed a three-dimensional dynamic time warping algorithm for online alignment of tapping events to scores, achieving 86% group-level accuracy on a small custom test set.
- Developing automatic melody recognition, an automatic-accompaniment benchmark, and online alignment of tapping events to score positions to automate score generation in PiCoMaestro, improve accompaniment expressivity, and support fine-grained polyphonic control.

Magic Quill: Instantaneous MIDI-to-Score Transcription

Dec. 2025 – Mar. 2026

- Designed a web-based instantaneous performance-MIDI-to-score system that runs online inference and offline refinement in parallel for responsive, accurate transcription.
- Proposed a causal LSTM-based symbolic beat and downbeat tracker optimized for commodity hardware, achieving sub-10 ms latency while retaining 83.9% performance relative to the state-of-the-art offline method.
- Introduced a Transformer-based offline beat phase tracker that provides high-throughput refinement with a compact metrical representation, achieving competitive performance with state-of-the-art performance-MIDI-to-score models.

Tap to Music

Nov. 2025 – May 2026

- Trained an LSTM-based model for low-latency online music generation conditioned on computer keyboard input, prior output, and control signals, such as register, chord, and musical score.
- Developed a web-based demo using ONNX Runtime and React, achieving under 15ms end-to-end latency.

PUBLICATIONS

- **L. Ye, R. Dannenberg, C. Donahue.** “Real-Time Performance-MIDI-to-Score Transcription with Causal Sequence Modeling and Latent Distillation”, *Manuscript in preparation*, 2026.
- **L. Ye, R. Dannenberg, C. Donahue.** “Magic Quill: Instantaneous MIDI-to-Score Transcription via Online Inference and Offline Refinement”, *Manuscript under review*, 2026.
- **L. Ye,** “Masked Expressiveness: Conditioned Generation of Piano Key Striking Velocity Using Masked Language Modeling”, *Highlights in Science, Engineering and Technology*, 120, 81–87. 2024.
- **L. Ye,** “Chord sense: Enhancing Stylistic Chord Progression Generation with Fine-tuned Transformers”, *Applied and Computational Engineering*, 68, 359-367. 2024.

PROFESSIONAL EXPERIENCE

Tencent Music Entertainment

Shenzhen, China

Research and Development Intern

Jun. 2026 – Aug. 2026

- Developed an audio-to-score transcription system, achieving state-of-the-art results as of 2026 on the MUSTER and MV2H evaluation metrics.
- Designed heuristic music-understanding algorithms for custom key, mode, and chord estimation tasks, improving accuracy by 34% over baseline methods.
- Designed heuristic pop song piano arrangement algorithms, improving the score difficulty factor by 18% while preserving the original arrangement’s musicality and flow.

Shenzhen Mango Future Technology Co., Ltd.

Shenzhen, China

Algorithm Engineer

Feb. 2022 – Aug. 2024

- Led development of a band accompaniment generation system serving 3,000 daily active users. Designed a chroma-based chord classifier with 90% weighted accuracy across 54 classes using support vector machines, implemented synthesizer logic, and coordinated client/server-side deployment.
- Refactored and extended a real-time score synthesis and playback system in C++, improving support for custom instruments and audio effect plugins. Improved synthesis quality through dynamic range control, room impulse response, and custom soundfonts for realistic piano and string sounds.
- Deployed deep learning models for expressive piano dynamic prediction and developed a sampling synthesis algorithm for solfège signing and interactive learning.

EXTRACURRICULAR ACTIVITIES

University of Rochester

Rochester, NY

Chamber Orchestra – Principal Clarinet

Sep. 2017 – May 2020

- Performed in university concerts and at local venues throughout the Rochester community.

University of Rochester

Rochester, NY

Chamber Orchestra – Soloist

Sep. 2019 – Feb. 2020

- The pianist winner of the University of Rochester River Campus 2019/2020 Concerto Competition. Received high praise from all judges.
- Performed John Field’s Piano Concerto No. 2 mvt. 1 with the University of Rochester Chamber Orchestra at the Strong Auditorium.

SKILLS

- Research: music information retrieval, audio signal processing, deep learning for music and audio, sequence models for real-time music processing, natural language processing
- Programming: Python, C++, Java, JavaScript/TypeScript
- Frameworks: PyTorch, TensorFlow.js, React, FluidSynth
- Languages: English (fluent), Japanese (intermediate), Chinese (native)

SELECTED COURSEWORK

• CMU10725 – Optimization for ML	Spring 2026
• CMU15622 – Intro to Computer Music	Spring 2026
• CMU57594 – Shaping Time in Performance	Spring 2026
• CMU10734 – Foundations of Autonomous Decision Making under Uncertainty	Fall 2025
• CMU10701 – Introduction to Machine Learning	Fall 2025
• CMU15798 – Generative AI for Music and Audio	Fall 2025
• URCSC261 – Intro to Databases	Spring 2020
• URCSC242 – Artificial Intelligence	Spring 2019
• URCSC240 – Data Mining	Fall 2019
• URCSC172 – Data Structures & Algorithms	Fall 2018
• URMUR221~224 – Music History	Fall 2018~Spring 2021
• URMUR211~212 – Music Theory	Fall 2018~Spring 2019
• URMUR114~115 – Musicianship	Spring 2019~Fall 2019
• URPA160 – Primary Piano	Fall 2018~Spring 2020